

# Policy gradient on the soccer game

REINFORCE, A2C and PPO, trained the standard way and by fictitious play, scored by one number: how often the trained player wins when the opponent is the best response to it.

## Summary

- Against the best response, the exact solver wins 67% of games on the random board; the six learners win 5% to 14%. On the deterministic board the exact solver ties itself, so it wins 0%.
- Against a random player the learners win 61% to 96% of games.
- Starting from the 94 states where the exact answer has to mix, they win 11% to 15% against the exact Nash policy; the exact solver wins 50%.
- Fictitious-play training wins more often against a random player; neither way of training reaches the exact solver's rate against the best response.

3 seeds per learner, 1,000 games per opponent.

## 1. How to read the numbers

| number                                 | what it is                                                                                                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Win rate against the best response     | The share of 1,000 games the trained player wins when the opponent plays the exact best response to it, the move that is best against exactly what our player will do. A player who always does the same thing loses every game; the exact solver mixes optimally, so its rate is the ceiling to compare with. |
| Win rate against the exact Nash policy | The same, against the exact equilibrium player.                                                                                                                                                                                                                                                                |
| Win rate against a random player       | A sanity check: every learner should beat a player who moves at random.                                                                                                                                                                                                                                        |
| A win                                  | The player scores while the opponent does not. On the first two boards the first goal ends the game; in the continuing game play restarts after a goal and a win is more goals than the opponent in 100 steps. Games with no goal, or equal goals, are ties.                                                   |

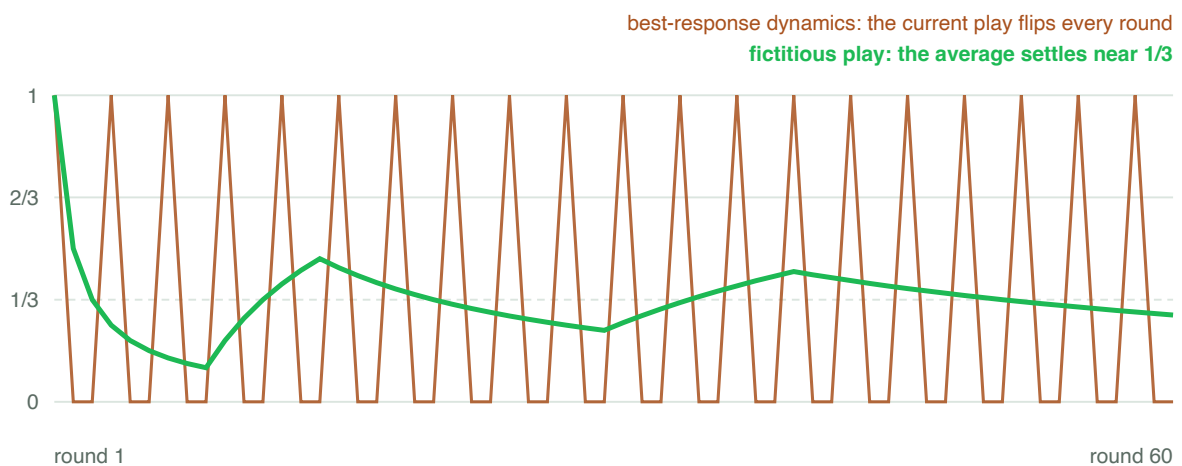
Exploitability, the best-response player's gain, is not reported: it measures the attacker, and it is the player we train that we want to judge.

## 2. How it is implemented

| setting         | value                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Game            | 7×5 board, goal rows 1–3. Fixed 100 steps, discount 0.9, and the number of steps left is a network input.                                                                                                                                                                                                                                               |
| Returns         | REINFORCE: the discounted return from each step to the end of the episode, no baseline. A2C: 10-step bootstrapped advantage from a learned critic. A2C with the exact critic: the critic is the exact solver's value, frozen, so the advantage is the exact $Q(s, a_0, a_1)$ minus $V(s)$ . PPO: GAE ( $\lambda$ 0.95), ratio clipped at 0.2, 4 epochs. |
| Updates         | Every iteration: 64 episodes of 100 steps, then one gradient step per player on all of that data (four for PPO). 2,000 iterations. Learning rate $10^{-3}$ , entropy bonus 0.01, 64×64 network, softmax output, a separate network for each player. No mini-batches, no replay.                                                                         |
| Standard        | Both players' current networks play each other and keep updating: real-time best responses.                                                                                                                                                                                                                                                             |
| Fictitious play | Each player trains against a frozen snapshot of the other, drawn at random from all its past snapshots (one every 5 iterations); the policy it reports is the average of its own snapshots.                                                                                                                                                             |

## 3. Why fictitious play

On rock-paper-scissors, answering the opponent's latest move makes the play flip between rock, paper and scissors for ever. Answering the average of everything it has played converges to one third each.

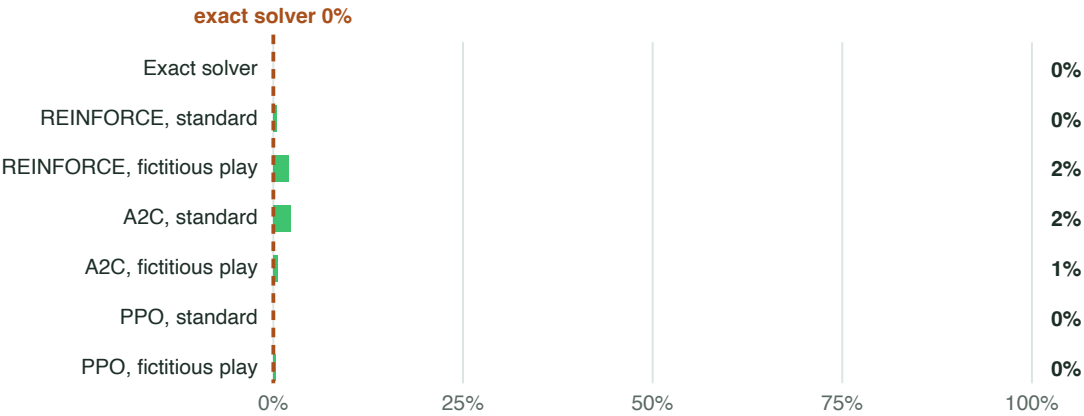


## 4. Results

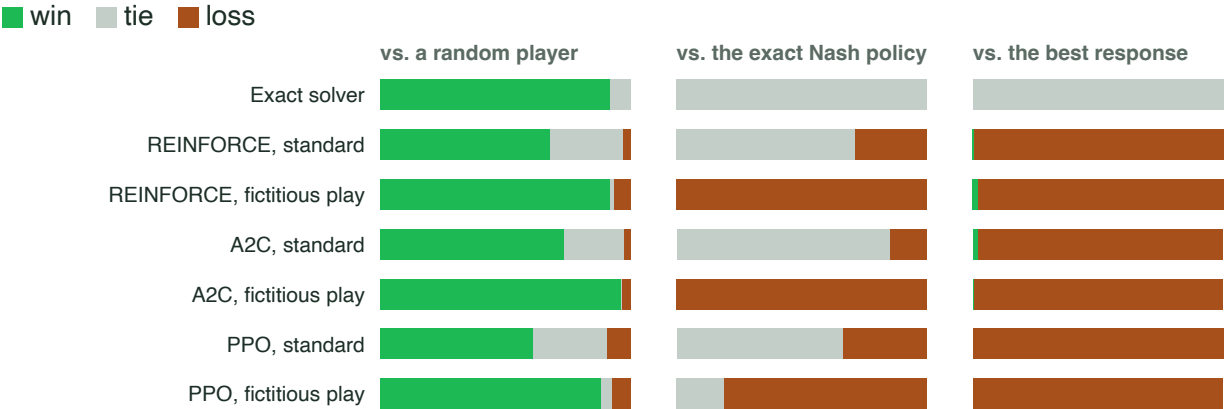
### The deterministic board (A10)

Players move simultaneously and the carrier wins every contested square, so the exact solution is pure everywhere and ties itself.

**Win rate against the best response** (mean of 3 seeds; dashed line: the exact solver). The exact solver wins 0%; the learners win 0% to 2%.



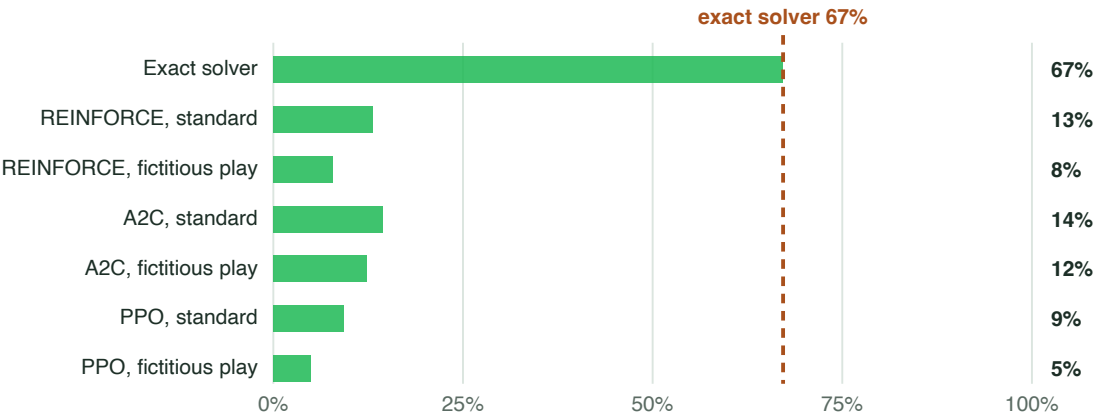
**Wins, ties and losses against each opponent.** Against the exact Nash policy the standard learners tie 67% to 85% of games and the fictitious-play learners 0% to 19%.



**The random move-order board**

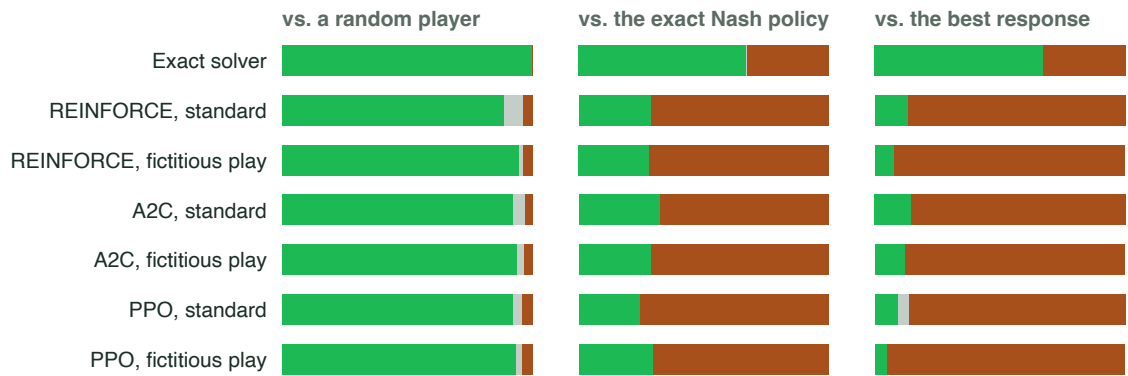
Moves are applied in a random order each step; the exact solution mixes at 94 stage games.

**Win rate against the best response** (mean of 3 seeds; dashed line: the exact solver). The exact solver wins 67%; the learners win 5% to 14%.



**Wins, ties and losses against each opponent.**

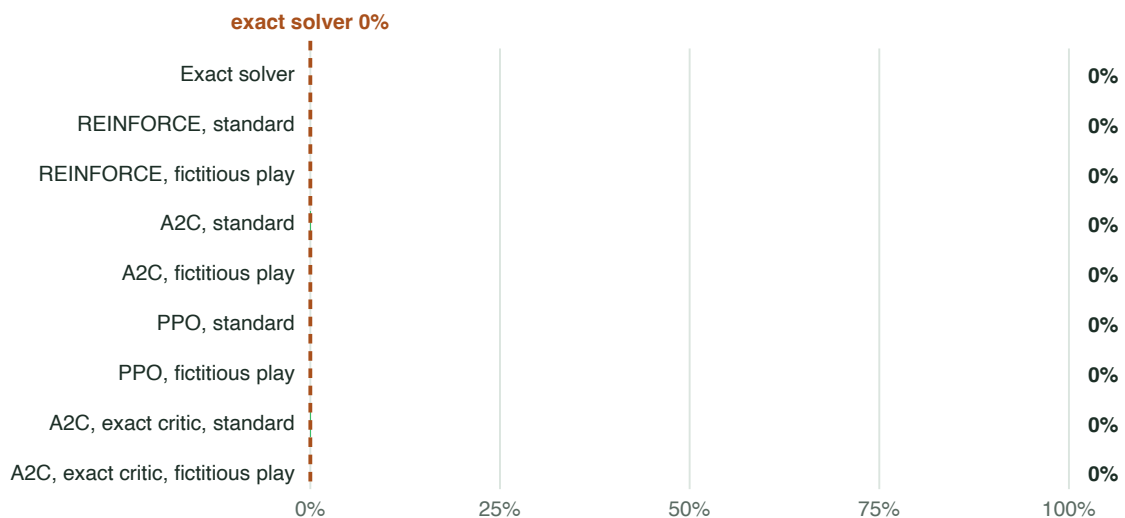
win tie loss



## The continuing game

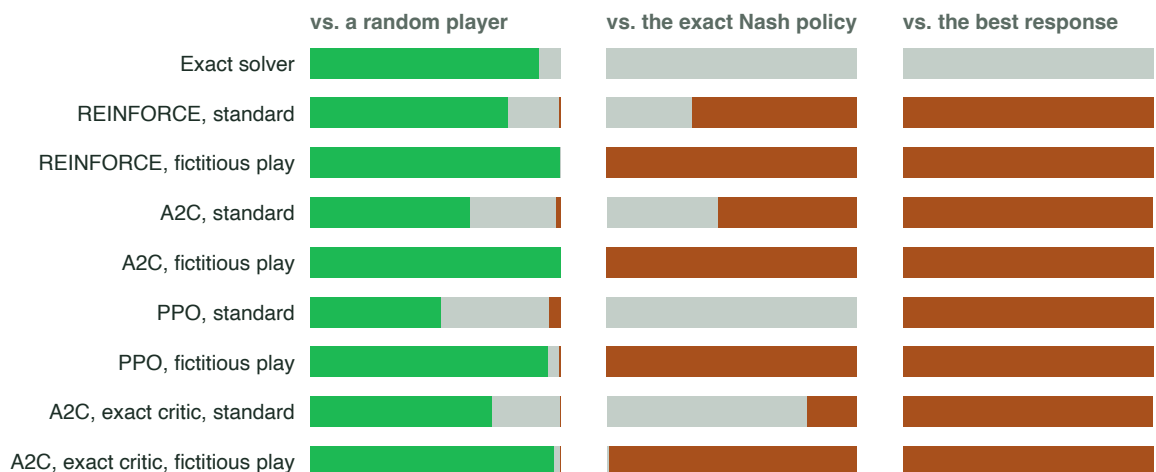
Play restarts after every goal and runs for exactly 100 steps; a win is more goals than the opponent. Also shows A2C with the exact solver's values as a frozen critic.

**Win rate against the best response** (mean of 3 seeds; dashed line: the exact solver). The exact solver wins 0%; the learners win 0% to 0%.



**Wins, ties and losses against each opponent.** Against the exact Nash policy the standard learners tie 34% to 100% of games and the fictitious-play learners 0% to 1%. A2C with the exact critic ties the exact Nash policy in 80% of games, against 45% for A2C with a learned critic.

■ win ■ tie ■ loss



## Win rate against the best response, all boards

| learner                    | deterministic board             | random board                    | continuing game                 |
|----------------------------|---------------------------------|---------------------------------|---------------------------------|
| Exact solver               | <b>0%</b><br>tie 100% · loss 0% | <b>67%</b><br>tie 0% · loss 33% | <b>0%</b><br>tie 100% · loss 0% |
| REINFORCE, standard        | <b>0%</b><br>tie 0% · loss 100% | <b>13%</b><br>tie 0% · loss 87% | <b>0%</b><br>tie 0% · loss 100% |
| REINFORCE, fictitious play | <b>2%</b><br>tie 0% · loss 98%  | <b>8%</b><br>tie 0% · loss 92%  | <b>0%</b><br>tie 0% · loss 100% |
| A2C, standard              | <b>2%</b><br>tie 0% · loss 98%  | <b>14%</b><br>tie 0% · loss 86% | <b>0%</b><br>tie 0% · loss 100% |
| A2C, fictitious play       | <b>1%</b><br>tie 0% · loss 99%  | <b>12%</b><br>tie 0% · loss 88% | <b>0%</b><br>tie 0% · loss 100% |
| PP0, standard              | <b>0%</b><br>tie 0% · loss 100% | <b>9%</b><br>tie 4% · loss 86%  | <b>0%</b><br>tie 0% · loss 100% |
| PP0, fictitious play       | <b>0%</b><br>tie 0% · loss 100% | <b>5%</b><br>tie 0% · loss 95%  | <b>0%</b><br>tie 0% · loss 100% |

## 5. Mixed states of the random board

At 94 states the exact equilibrium has to randomize. Each learner starts 20 games from every one of them. Distance is the total-variation distance between the learner's probabilities and the exact mix at the first step (0 = identical, 1 = no overlap); "player 0 mixes" is the share of states where it puts less than 90% on its top move.

| learner                    | wins vs. exact Nash             | wins vs. best response          | distance from the exact mix | player 0 mixes |
|----------------------------|---------------------------------|---------------------------------|-----------------------------|----------------|
| Exact solver               | <b>50%</b><br>tie 0% · loss 50% | <b>50%</b><br>tie 0% · loss 50% | 0                           |                |
| REINFORCE, standard        | <b>12%</b><br>tie 0% · loss 88% | <b>16%</b><br>tie 0% · loss 84% | 0.64                        | 72%            |
| REINFORCE, fictitious play | <b>11%</b><br>tie 0% · loss 89% | <b>23%</b><br>tie 0% · loss 77% | 0.65                        | 98%            |
| A2C, standard              | <b>14%</b><br>tie 0% · loss 86% | <b>18%</b><br>tie 0% · loss 82% | 0.59                        | 71%            |
| A2C, fictitious play       | <b>11%</b><br>tie 0% · loss 89% | <b>20%</b><br>tie 0% · loss 80% | 0.65                        | 100%           |
| PP0, standard              | <b>15%</b><br>tie 0% · loss 85% | <b>25%</b><br>tie 0% · loss 75% | 0.67                        | 29%            |
| PP0, fictitious play       | <b>14%</b><br>tie 0% · loss 86% | <b>40%</b><br>tie 0% · loss 60% | 0.64                        | 72%            |

### Three of these states in full

**Mixed state 1 (three-way mix): player 0 at (0, 0), player 1 at (2, 0), player 0 has the ball**

| policy                     | player 0   |    |            |            | player 1   |    |            |     |
|----------------------------|------------|----|------------|------------|------------|----|------------|-----|
|                            | U          | D  | L          | R          | U          | D  | L          | R   |
| Exact solver               | 43%        | 0% | <b>55%</b> | 2%         | <b>92%</b> | 6% | 2%         | 0%  |
| REINFORCE, standard        | <b>89%</b> | 0% | 0%         | 11%        | <b>83%</b> | 0% | 3%         | 14% |
| REINFORCE, fictitious play | <b>77%</b> | 2% | 1%         | 21%        | 9%         | 2% | <b>86%</b> | 3%  |
| A2C, standard              | <b>66%</b> | 0% | 0%         | 34%        | <b>88%</b> | 0% | 8%         | 4%  |
| A2C, fictitious play       | <b>60%</b> | 2% | 1%         | 37%        | <b>54%</b> | 1% | 40%        | 4%  |
| PP0, standard              | <b>77%</b> | 0% | 0%         | 23%        | <b>99%</b> | 0% | 0%         | 1%  |
| PP0, fictitious play       | 22%        | 0% | 0%         | <b>77%</b> | 1%         | 1% | <b>99%</b> | 0%  |

**Mixed state 2 (two-way mix): player 0 at (0, 0), player 1 at (1, 1), player 0 has the ball**

| policy                     | player 0   |    |            |            | player 1 |            |            |            |
|----------------------------|------------|----|------------|------------|----------|------------|------------|------------|
|                            | U          | D  | L          | R          | U        | D          | L          | R          |
| Exact solver               | 5%         | 0% | <b>95%</b> | 0%         | 0%       | <b>95%</b> | 5%         | 0%         |
| REINFORCE, standard        | <b>70%</b> | 0% | 0%         | 30%        | 0%       | 21%        | 31%        | <b>48%</b> |
| REINFORCE, fictitious play | <b>76%</b> | 2% | 1%         | 21%        | 2%       | 2%         | <b>92%</b> | 3%         |
| A2C, standard              | <b>51%</b> | 0% | 0%         | 49%        | 0%       | 5%         | 27%        | <b>68%</b> |
| A2C, fictitious play       | 48%        | 2% | 1%         | <b>49%</b> | 3%       | 12%        | <b>61%</b> | 24%        |
| PP0, standard              | 9%         | 0% | 0%         | <b>91%</b> | 0%       | 0%         | 0%         | <b>99%</b> |
| PP0, fictitious play       | 2%         | 0% | 0%         | <b>98%</b> | 0%       | <b>91%</b> | 9%         | 0%         |

**Mixed state 3 (two-way mix): player 0 at (0, 1), player 1 at (1, 0), player 1 has the ball**

| policy                     | player 0   |            |     |            | player 1   |            |             |     |
|----------------------------|------------|------------|-----|------------|------------|------------|-------------|-----|
|                            | U          | D          | L   | R          | U          | D          | L           | R   |
| Exact solver               | 0%         | 0%         | 39% | <b>61%</b> | 0%         | <b>68%</b> | 0%          | 32% |
| REINFORCE, standard        | <b>62%</b> | 0%         | 29% | 8%         | <b>99%</b> | 0%         | 1%          | 0%  |
| REINFORCE, fictitious play | <b>68%</b> | 2%         | 1%  | 28%        | 15%        | 4%         | <b>79%</b>  | 2%  |
| A2C, standard              | <b>42%</b> | 0%         | 32% | 26%        | <b>98%</b> | 0%         | 2%          | 0%  |
| A2C, fictitious play       | 4%         | <b>40%</b> | 40% | 15%        | 5%         | 2%         | <b>92%</b>  | 1%  |
| PP0, standard              | 2%         | 0%         | 3%  | <b>95%</b> | 0%         | 0%         | <b>100%</b> | 0%  |
| PP0, fictitious play       | <b>88%</b> | 0%         | 0%  | 11%        | <b>96%</b> | 0%         | 3%          | 0%  |

## 6. Action probabilities

The probability each trained network gives to every move, at four fixed positions of the deterministic board (seed 0), next to the exact solver's move. The most likely move is in bold.

Kickoff: player 0 at (0, 1), player 1 at (6, 3), player 0 has the ball

| policy                     | player 0 |     |     |     | player 1 |      |     |     |
|----------------------------|----------|-----|-----|-----|----------|------|-----|-----|
|                            | U        | D   | L   | R   | U        | D    | L   | R   |
| Exact solver               | 100%     | 0%  | 0%  | 0%  | 100%     | 0%   | 0%  | 0%  |
| REINFORCE, standard        | 46%      | 22% | 31% | 0%  | 7%       | 1%   | 57% | 35% |
| REINFORCE, fictitious play | 41%      | 1%  | 29% | 29% | 1%       | 17%  | 82% | 1%  |
| A2C, standard              | 31%      | 26% | 33% | 10% | 9%       | 1%   | 69% | 21% |
| A2C, fictitious play       | 62%      | 0%  | 13% | 25% | 1%       | 8%   | 90% | 1%  |
| PP0, standard              | 100%     | 0%  | 0%  | 0%  | 0%       | 100% | 0%  | 0%  |
| PP0, fictitious play       | 93%      | 0%  | 0%  | 7%  | 0%       | 3%   | 96% | 0%  |

Players side by side in a goal row: player 0 at (3, 2), player 1 at (4, 2), player 0 has the ball

| policy                     | player 0 |     |     |      | player 1 |     |     |      |
|----------------------------|----------|-----|-----|------|----------|-----|-----|------|
|                            | U        | D   | L   | R    | U        | D   | L   | R    |
| Exact solver               | 100%     | 0%  | 0%  | 0%   | 0%       | 0%  | 0%  | 100% |
| REINFORCE, standard        | 88%      | 3%  | 6%  | 2%   | 4%       | 31% | 31% | 34%  |
| REINFORCE, fictitious play | 5%       | 1%  | 2%  | 92%  | 25%      | 32% | 42% | 1%   |
| A2C, standard              | 26%      | 36% | 28% | 10%  | 7%       | 5%  | 18% | 69%  |
| A2C, fictitious play       | 20%      | 1%  | 3%  | 76%  | 37%      | 57% | 5%  | 1%   |
| PP0, standard              | 0%       | 0%  | 0%  | 100% | 0%       | 0%  | 0%  | 100% |
| PP0, fictitious play       | 79%      | 0%  | 0%  | 21%  | 35%      | 5%  | 59% | 0%   |

Carrier near the goal, defender far away: player 0 at (5, 1), player 1 at (6, 3), player 0 has the ball

| policy                     | player 0 |     |     |      | player 1 |      |    |    |
|----------------------------|----------|-----|-----|------|----------|------|----|----|
|                            | U        | D   | L   | R    | U        | D    | L  | R  |
| Exact solver               | 0%       | 0%  | 0%  | 100% | 0%       | 100% | 0% | 0% |
| REINFORCE, standard        | 90%      | 4%  | 5%  | 1%   | 0%       | 100% | 0% | 0% |
| REINFORCE, fictitious play | 11%      | 1%  | 3%  | 86%  | 1%       | 95%  | 3% | 1% |
| A2C, standard              | 25%      | 39% | 25% | 11%  | 0%       | 97%  | 0% | 3% |
| A2C, fictitious play       | 31%      | 0%  | 4%  | 65%  | 1%       | 95%  | 3% | 1% |
| PP0, standard              | 0%       | 0%  | 0%  | 100% | 0%       | 100% | 0% | 0% |
| PP0, fictitious play       | 91%      | 0%  | 0%  | 9%   | 0%       | 99%  | 1% | 0% |

The right player holds the ball, players stacked: player 0 at (3, 2), player 1 at (3, 3), player 1 has the ball

| policy                     | player 0    |    |            |             | player 1 |             |             |    |
|----------------------------|-------------|----|------------|-------------|----------|-------------|-------------|----|
|                            | U           | D  | L          | R           | U        | D           | L           | R  |
| Exact solver               | <b>100%</b> | 0% | 0%         | 0%          | 0%       | 0%          | <b>100%</b> | 0% |
| REINFORCE, standard        | 3%          | 3% | <b>91%</b> | 3%          | 0%       | 0%          | <b>99%</b>  | 1% |
| REINFORCE, fictitious play | 14%         | 2% | 15%        | <b>69%</b>  | 1%       | 2%          | <b>96%</b>  | 0% |
| A2C, standard              | 4%          | 1% | <b>71%</b> | 24%         | 0%       | 14%         | <b>84%</b>  | 1% |
| A2C, fictitious play       | 6%          | 2% | 38%        | <b>53%</b>  | 1%       | 2%          | <b>97%</b>  | 0% |
| PP0, standard              | 0%          | 0% | 0%         | <b>100%</b> | 0%       | <b>100%</b> | 0%          | 0% |
| PP0, fictitious play       | 6%          | 0% | 0%         | <b>94%</b>  | 0%       | 0%          | <b>99%</b>  | 0% |

## 7. More training

Standard training at 8,000 instead of 2,000 iterations (two seeds each), deterministic board.

| learner                | wins vs. exact Nash,<br>2,000 iterations | wins vs. exact Nash,<br>8,000 iterations | wins vs. random,<br>8,000 iterations |
|------------------------|------------------------------------------|------------------------------------------|--------------------------------------|
| REINFORCE,<br>standard | <b>0%</b><br>tie 71% · loss 29%          | <b>0%</b><br>tie 38% · loss 62%          | <b>91%</b><br>tie 8% · loss 2%       |
| A2C, standard          | <b>0%</b><br>tie 85% · loss 15%          | <b>0%</b><br>tie 79% · loss 21%          | <b>79%</b><br>tie 20% · loss 1%      |
| PP0, standard          | <b>0%</b><br>tie 67% · loss 33%          | <b>0%</b><br>tie 0% · loss 100%          | <b>97%</b><br>tie 0% · loss 2%       |

## 8. Fictitious play with long best responses

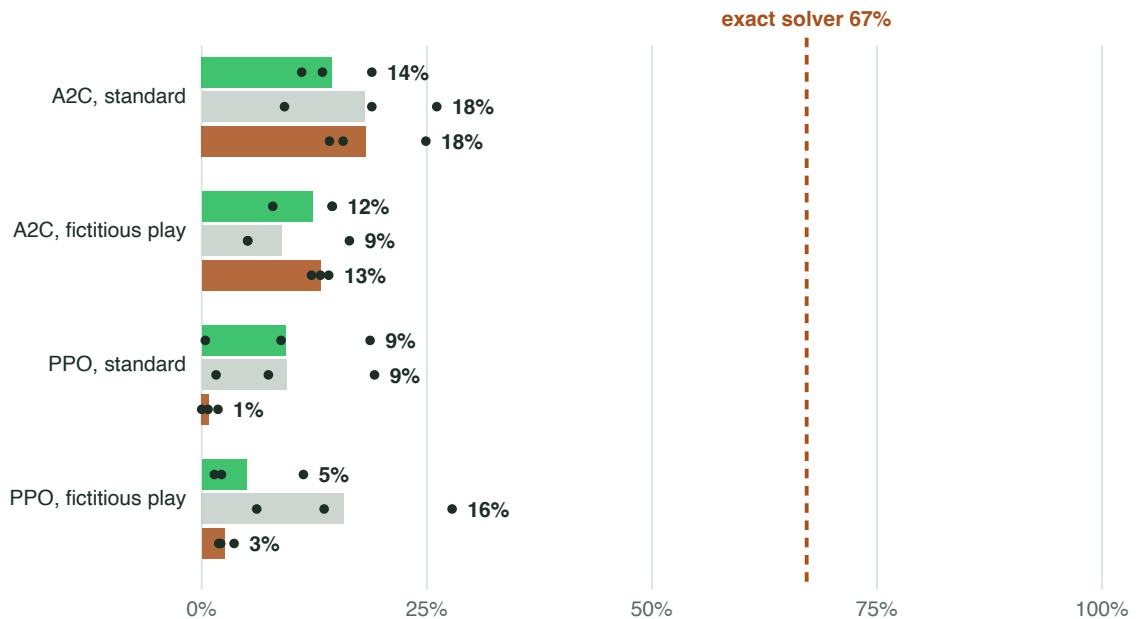
Each round a player trains for 100 (or 300) iterations against the average of its opponent's earlier best responses, then adds the result to its history. Win rate against the best response at each checkpoint round (one seed per run).

| run                                         | round<br>5 | round<br>10 | round<br>15 | round<br>20 | round<br>25 | round<br>30 |
|---------------------------------------------|------------|-------------|-------------|-------------|-------------|-------------|
| A2C, 100 iterations per best response       | 0%         | 1%          | 2%          | 2%          | 2%          | 1%          |
| A2C, 300 iterations per best response       | 1%         | 0%          | 0%          |             |             |             |
| PP0, 100 iterations per best response       | 0%         | 0%          | 0%          | 0%          | 0%          | 0%          |
| REINFORCE, 100 iterations per best response | 0%         | 0%          | 1%          | 1%          | 1%          | 1%          |

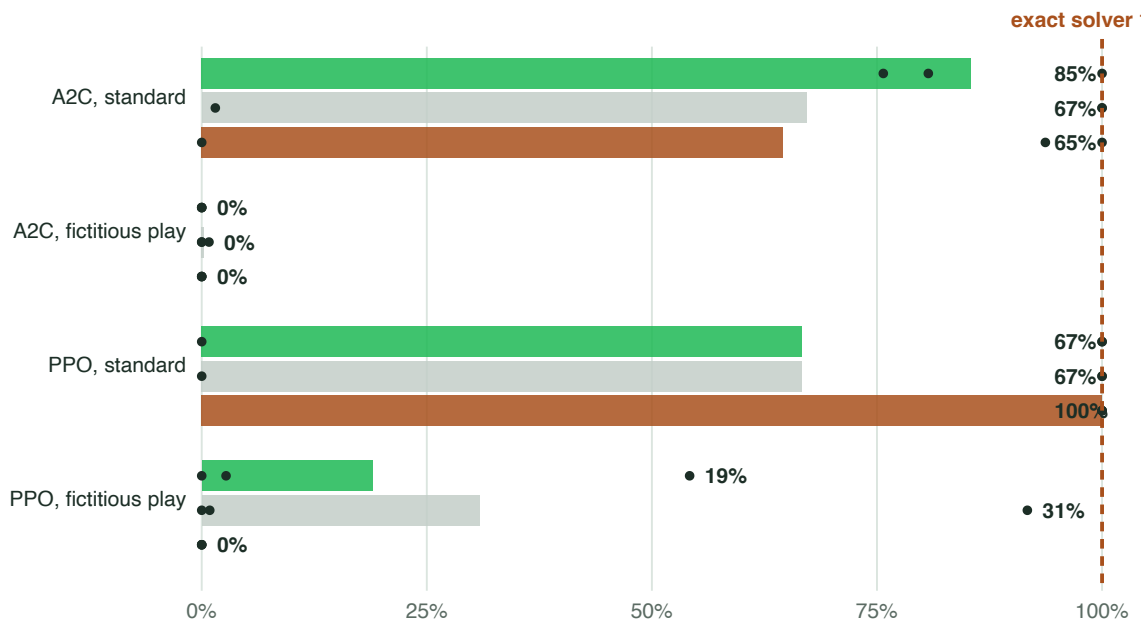
## 9. Network sharing and trimming

A2C and PPO, three seeds each, on all three boards. **Shared network:** both players' policies are two output slices of one network, so each player's update also moves the other. **Last 10 steps left out:** the final 10 of the 100 steps of every episode are not used in the loss, though they still count in the returns of earlier steps. Bars are the mean over seeds; dots are single seeds. ■ separate networks, ■ one shared network, ■ last 10 steps left out.

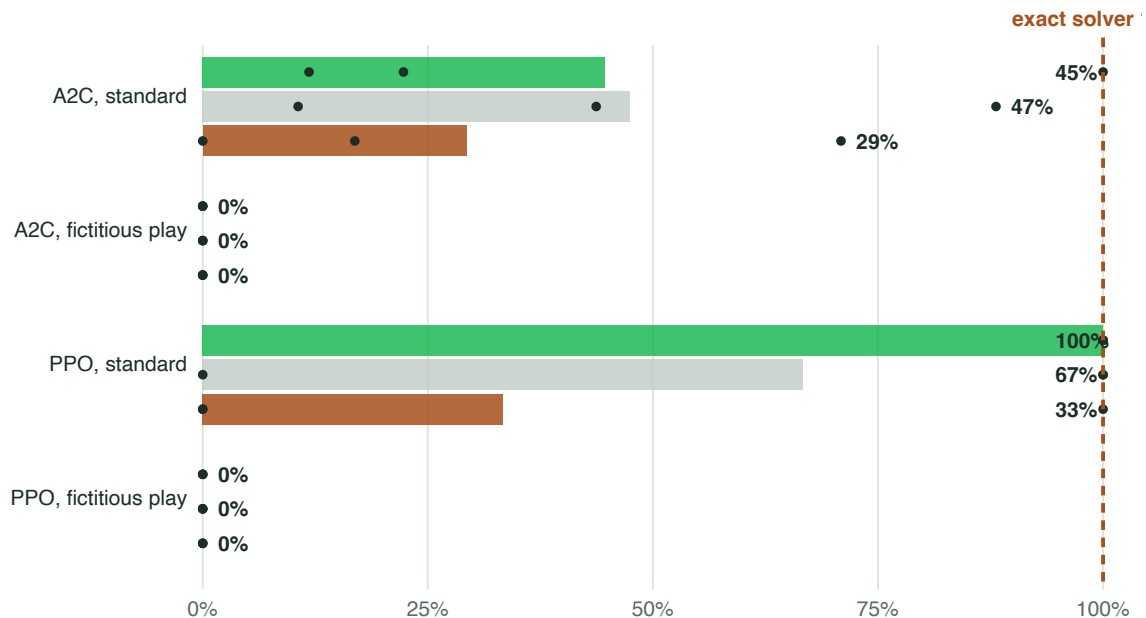
### Random move-order board: win rate against the best response



### Deterministic board: games tied with the exact Nash policy



### Continuing game: games tied with the exact Nash policy



Random board: against separate networks, a shared network changes the win rate by +4, -3, +0, +11 points (A2C standard, A2C fictitious play, PPO standard, PPO fictitious play); single seeds of the same setup differ by up to 22 points. Leaving out the last 10 steps changes A2C by +4 and +1 points, but PPO with standard training stops scoring: it wins 19% of games against a random player, against 92% before. No variant gets close to the exact solver.

Deterministic board: no variant wins more than 2% of games against the best response, as for the baseline (the exact solver wins 0%, since it ties itself). The share of games tied with the exact Nash policy changes by -18, +0, +0, +12 points with a shared network and by -21, +0, +33, -19 points with the last 10 steps left out (A2C standard, A2C fictitious play, PPO standard, PPO fictitious play); single seeds of the same setup differ by up to 100 points. A tie does not mean equilibrium play: PPO with standard training ties all games once the last 10 steps are left out, but wins only 36% of games against a random player, against 61% before.

Continuing game: no variant wins more than 0% of games against the best response, as for the baseline. The share of games tied with the exact Nash policy changes by +3, +0, -33, +0 points with a shared network and by -15, +0, -67, +0 points with the last 10 steps left out (A2C standard, A2C fictitious play, PPO standard, PPO fictitious play); single seeds of the same setup differ by up to 100 points, so three seeds cannot settle changes of this size.